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**B. Tech. Degree VI Semester Special Supplementary Examination
November 2014**

**ME 601 INSTRUMENTATION AND CONTROL SYSTEMS
(2006 Scheme)**

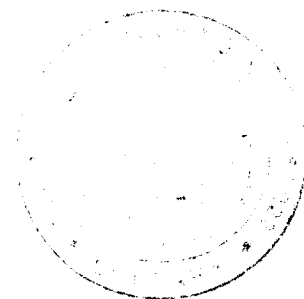
Time: 3 Hours

Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) By considering linear potentiometer, explain a zero order instrument.
 (b) Obtain the step response of a first order instrument.
 (c) Explain the working of a Gieger Muller counter, with a neat sketch.
 (d) Explain clearly the difference between bonded and unbonded strain gauge.
 (e) Explain any 3 time domain specifications.
 (f) Explain liquid level systems, with the help of an example.
 (g) State Routh-Hurwitz criterion for determination of stability.
 (h) Explain stepper motor, with the help of its operating principle.



PART B

(4 x 15 = 60)

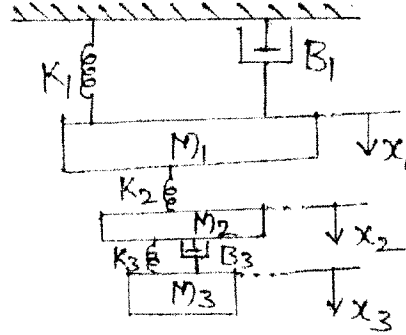
- II. (a) Explain the functional elements of a measuring system, with the help of an example. (10)
 (b) Explain desired, interfering, modifying inputs, with the help of an example. (5)
- OR**
- III. Calculate the time constant of a first order mercury in glass thermometer inside diameter of bulb 4 mm, assuming the bulb to be spherical, density of mercury = 13600 Kg/m³, specific heat = 0.15 KJ/Kg°C, heat transfer coefficient 40 W/m² °C if the thermometer bulb had been cylindrical in shape having the same volume and diameter as the spherical bulb. What would be the time constant (i) neglecting both the ends area of cylinder (ii) accounting for one of the end areas? (15)
- IV. (a) Explain optical pyrometer, with a neat sketch. (8)
 (b) Explain hydraulic dynamometer, with a neat sketch. (7)
- OR**
- V. (a) Explain the working of a ORSAT'S apparatus with a neat sketch. (10)
 (b) Explain, with block diagram, the working of a microphone. (5)

(P.T.O.)

- VI. The response of a servo mechanism if $C(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$ when subjected to a unit step input. Obtain an expression for closed loop transfer function. Determine the un-damped natural frequency and damping ratio. (15)

OR

- VII. Write the differential equation governing the mechanical system shown in the figure below. Draw the force-voltage and force current electrical analogous circuits and verify by writing mesh and node equations. (15)



- VIII. Sketch bode plot for the following transfer function and determine the system gain 'K' for the gain cross over frequency to be 5 rad/sec. (15)

$$G(s) = \frac{Ks^2}{(1+0.2s)(1+0.02s)}$$

OR

- IX. Sketch the root locus of the system whose open transfer function is (15)

$$G(s) = \frac{k}{s(s+2)(s+4)}$$

Find the value of 'K' so that the damping ratio of the closed loop system is 0.5.
